

Intelligence

Lecture 10

What intelligence tests omit

Dr Steven Poulter

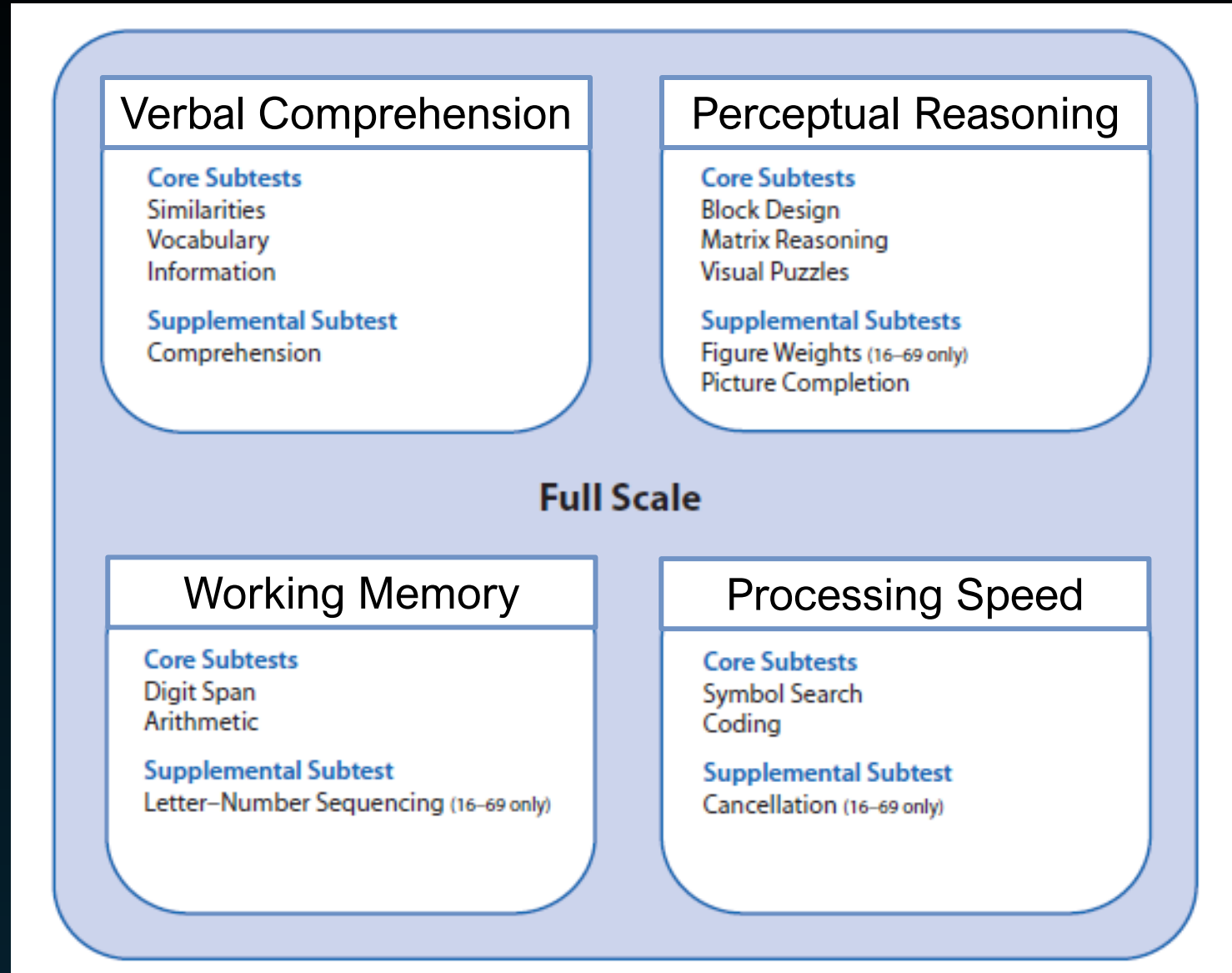
What IQ Tests Actually Measure

- Most IQ tests (e.g. WAIS, Raven's, etc.) are designed to measure:
 - > **General cognitive ability (g)** – shared variance across tasks.
 - > Subdomains: e.g. **verbal comprehension, working memory, processing speed, visuospatial reasoning.**
- They use **structured, often time-limited, decontextualised tasks** under standardised conditions.
- They aim for **reliability** and **predictive validity** (e.g. school performance, some job performance).

Role of intelligence tests

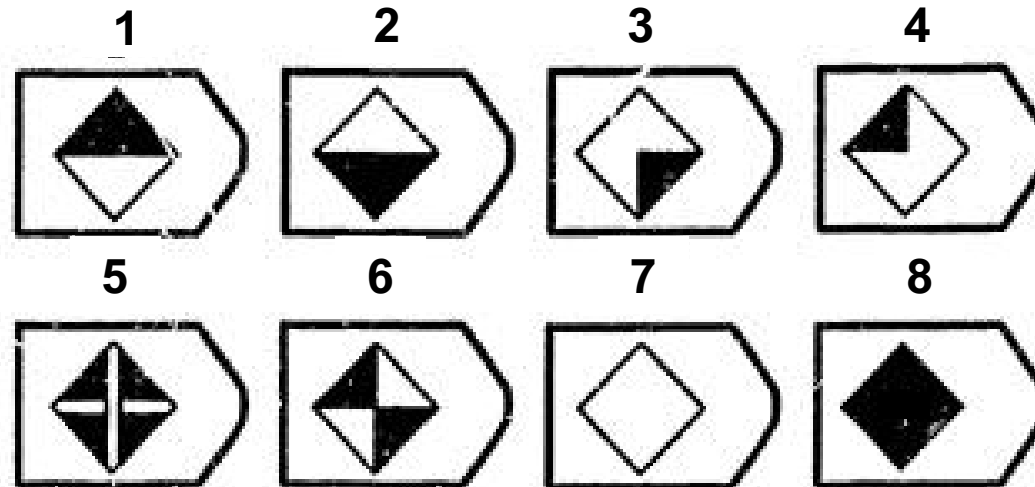
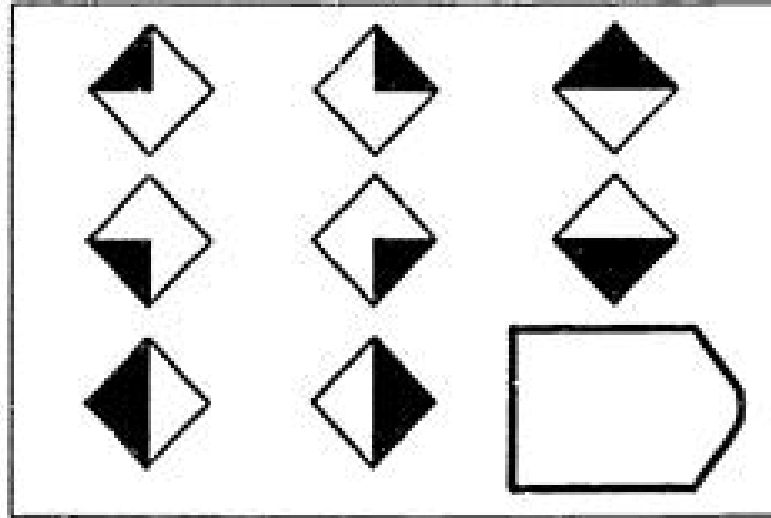
- To assess a full range of abilities, needs to contain a number of tasks
- Mainly used for **selection** (streaming children in schools, job recruitment), **diagnosis** (learning difficulties, brain trauma), **prediction** (various life outcomes)

The Wechsler Adult Intelligence Scale IV



Raven's Progressive Matrices

Figuring out the **underlying rules** that explain the progression of shapes.



What intelligence tests omit

Emotional and social intelligence

IQ tests **don't directly measure:**

- Emotional awareness and regulation.
- Empathy and social perception.
- Skills in managing relationships and conflict.



Social and emotional skills are central to:

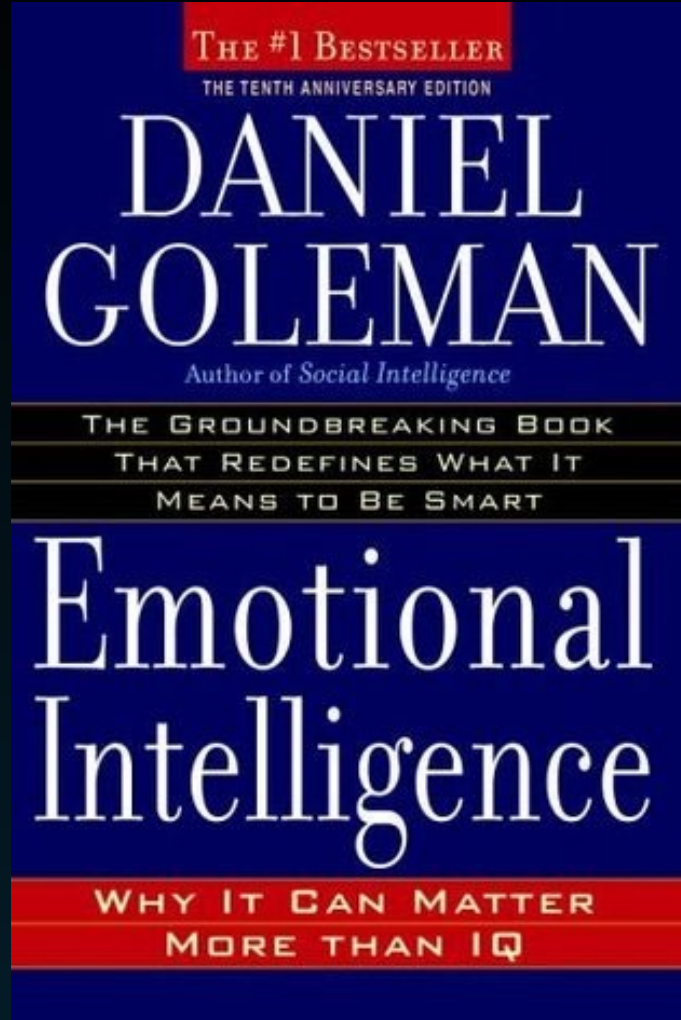
- Leadership and teamwork.
- Mental health.
- Relationship quality.

What intelligence tests omit

- ***Emotional Intelligence***: Ability to sense/evaluate others' emotions



EMOTIONAL INTELLIGENCE The ability to empathize and evaluate others' emotions.



- Popular book generated flood of interest in role of emotional intelligence in life
- Goleman surveyed years of research on psychological functioning and interpersonal skills
- ‘Soft-skills’ claimed to be crucial and popularised among educators, business and media

- ***Emotional Intelligence***: Ability to sense/evaluate others' emotions

Indicate how much of each emotion is expressed by this face:



No Happiness	1	2	3	4	5	Extreme Happiness
No Anger	1	2	3	4	5	Extreme Anger
No Fear	1	2	3	4	5	Extreme Fear

Example of the Multi-factor Emotional Intelligence Test (MEIS)/Mayer–Salovey–Caruso Emotional Intelligence Test (MSCEIT)



EQ – a set of skills for a complex world: personal, social, sensitivity, ‘common sense’

Also, ability to read political and social environments

Emotional Quotient vs Intelligence quotient

- Conflicting reports on the correlation between EQ and IQ (few reports but some positive, some negative and some report no correlation)

Emotional Intelligence: An Integrative Meta-Analysis and Cascading Model

Dana L. Joseph and Daniel A. Newman
University of Illinois at Urbana–Champaign

- Meta-analyses show certain **emotional intelligence (EI)** measures **predict job performance and work attitudes** (e.g. job satisfaction) even after controlling for IQ and Big Five traits, though **effects vary by how EI is measured**
- Mainly **small to moderate correlations** between EI and job performance / job attitudes

Emotional intelligence predicts life skills, but not as well as personality and cognitive abilities

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University of Adelaide, Adelaide, SA 5005, Australia

Received 15 May 2004; accepted 12 April 2005

Available online 5 July 2005

- 246 predominantly first-year tertiary students investigated relationships between EI and a number of life skills (academic achievement, life satisfaction, anxiety, problem-solving and coping).
- After controlling for personality and cognitive abilities, EI not a good predictor of 'life skills'

Enhanced emotional intelligence and success

- EI is a popular concept among general public – thinking of someone as a ‘people person’
- Some evidence that social skills contribute to success at work but typically if that individual is already competent at their job

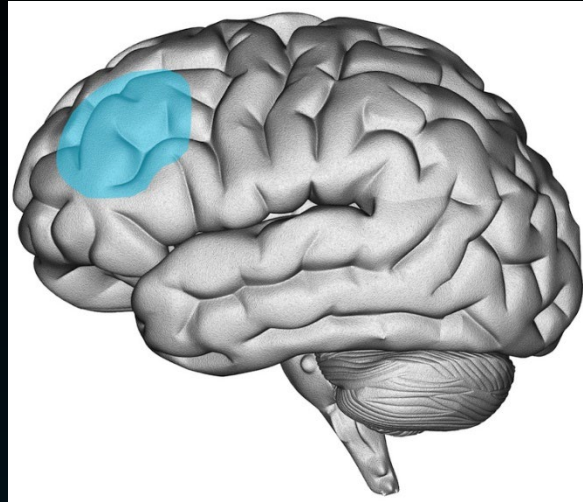
‘It remains uncertain whether there is anything about emotional intelligence that psychologists working within the fields of personality, intelligence, and applied psychological research do not already know’ (Roberts et al., 2001, p. 200).

What intelligence tests omit

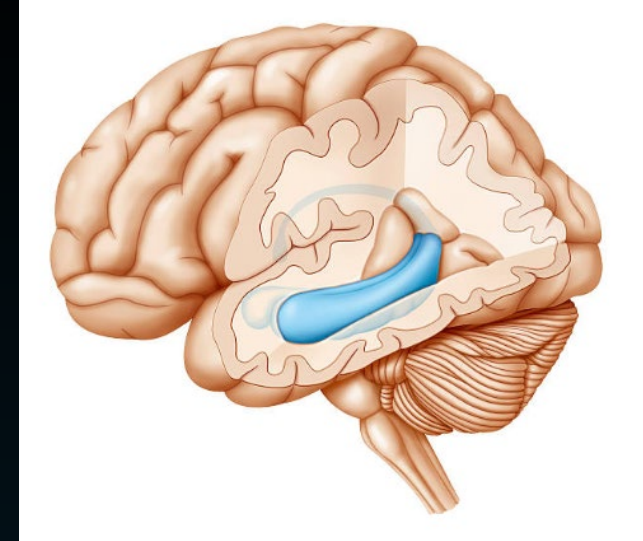
- **Wisdom:** Life skills and good common sense (similar to Sternberg's practical intelligence)



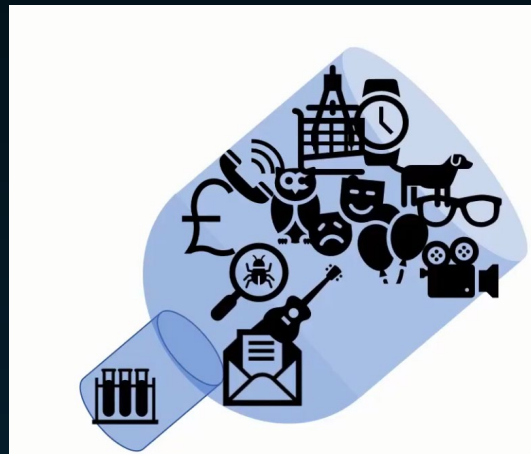
EFFECTS OF STRESS ON LEARNING



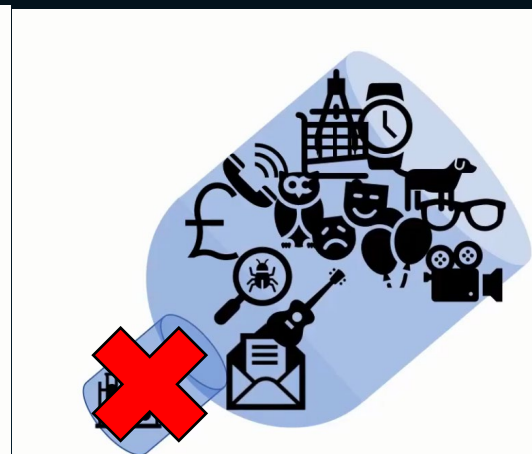
Dorsolateral Prefrontal Cortex
Working Memory



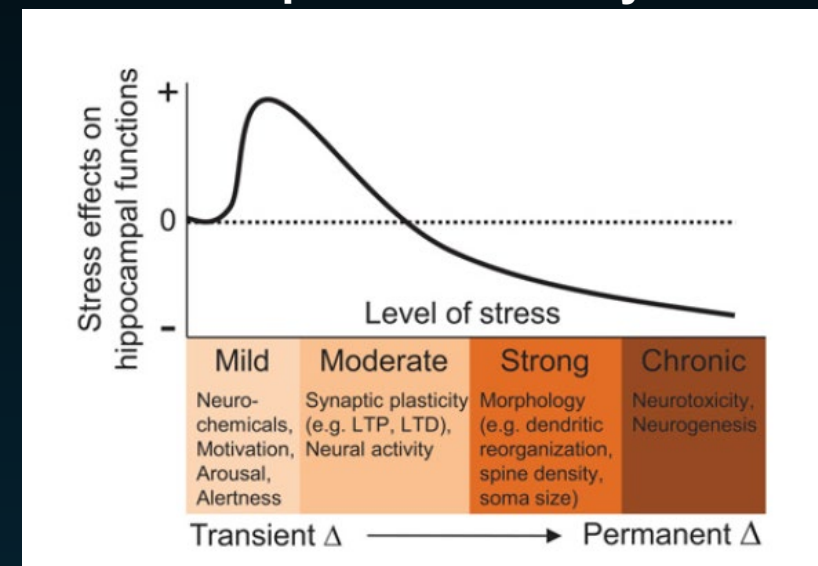
Hippocampus
Episodic Memory



Fales et al. 2010



Stress & Anxiety



Kim et al. 2015

What intelligence tests omit

Creativity



- Producing ideas that are both **novel** and **useful**.
- IQ tests mainly tap convergent thinking (one correct answer), whereas creativity often involves divergent thinking (many unusual answers).
- Review work arguing that:
 - >Creativity and IQ are moderately correlated at best; beyond a certain IQ threshold, the correlation is weak (e.g. Kim 2005, Kaufman & Plucker, 2011)
 - >IQ tests, even when they look like they involve creativity, don't actually capture real creative output in art, science, entrepreneurship (Kaufman, 2015).

What intelligence tests omit

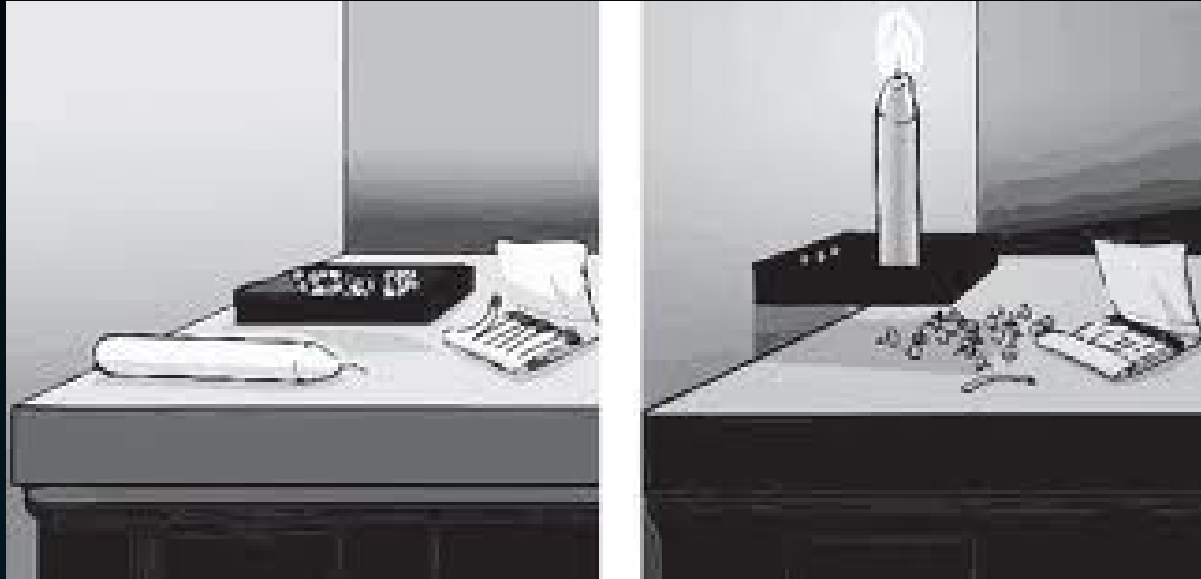
- ***Creativity***: Artistic or scientific ingenuity and ability to generate ideas (e.g. Duncker Candle Problem)



The solution is considered a measure of insight creativity when it involves the ability to see objects as performing different functions from what is typical.

What intelligence tests omit

- ***Creativity***: Artistic or scientific ingenuity and ability to generate ideas (e.g. Duncker Candle Problem)



The solution is considered a measure of insight creativity when it involves the ability to see objects as performing different functions from what is typical.

→ The solution is inconsistent with preexisting associations and expectations

What intelligence tests omit

Cultural context and test bias

- IQ tests have been criticised as culturally loaded and reflecting Western, schooled definitions of intelligence.
- Abilities valued in some cultures (e.g. social responsibility, community wisdom, ecological knowledge) may not show up in IQ test scores.
- Sternberg's cross-cultural work: children may perform poorly on Western-style tests but show high levels of practical intelligence in everyday tasks in their own context.



What intelligence tests omit

- ***Personality & Motivation***

- IQ is only part of the story for academic and life outcomes.
- Personality traits, especially **conscientiousness** (organisation, diligence, self-discipline), are robust predictors of school and university performance (alongside intelligence), e.g. Dumfart & Neubauer, 2016
- Meta-analytic and large-scale studies:
 - >Conscientiousness often rivals or exceeds IQ in predicting grades, particularly when content is less abstract.
 - >Other traits (e.g. openness to experience) also matter, but effects are smaller or depend on domain, e.g. Verbree et al. 2021
 - >Broader non-cognitive factors: Grit, self-efficacy, time management, and low test anxiety predict GPA, especially in diverse and nontraditional student populations, e.g. Cho & Serrano 2020
- IQ tests **do not assess** these characteristics.



Revision Session....

**Looking back over end of Intelligence
block MCQ test**